AI-BASED MONITORING AND CONTROL SYSTEM FOR IKAN PUYU GROWTH

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Introduction

- Traditional fish farming (especially for beginners) is inconsistent and prone to unpredictable yields and financial losses due to its manual, experience-based nature.
- A new system has been developed to predict the growth of climbing perch.
- This system also provides guidance on manipulations and monitoring needed to ensure stable and controlled fish growth.

Objectives

- To develop an AI-based system to monitor ikan puyu growth parameters.
- To implement automated control for optimal fish growth conditions.
- To predict ikan puyu growth trends using data analysis.

Problem Statements

Current Ikan Puyu farming methods face several challenges including:
Lack of real-time monitoring of critical parameters
Inefficient feeding management
Limited data analysis for growth prediction.
This leads to suboptimal growth performance
resource wastage, and reduced profitability.

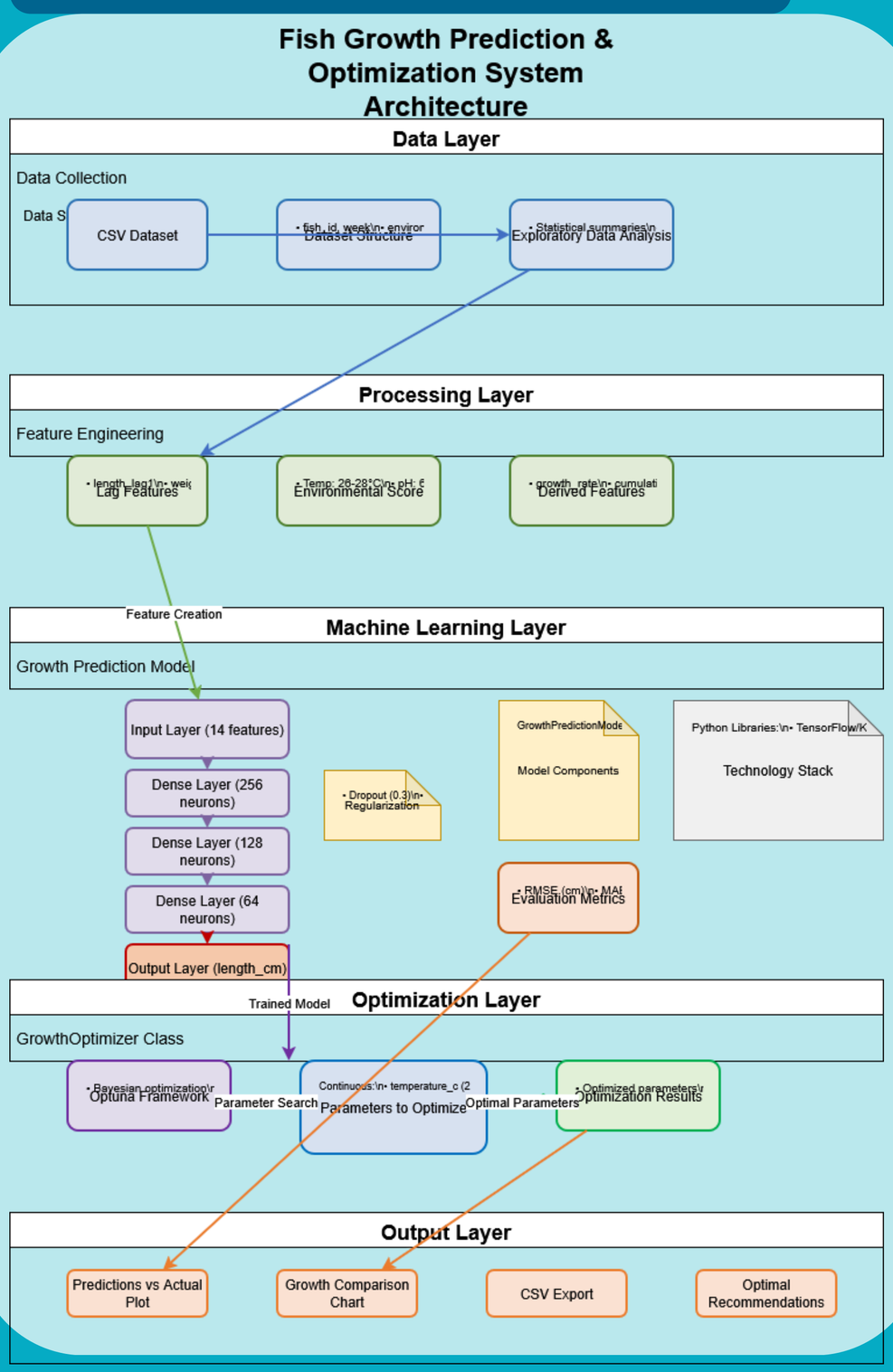
Methodology

- **Data Collection**: Environmental data (pH, temperature, turbidity, water level) is collected using sensors connected to Raspberry Pi.
- **AI Decision Logic**: Rule-based intelligence is applied based on expert knowledge to determine optimal pond conditions and trigger automated actions (feeding, water circulation).
- **System Implementation**: The system integrates sensors, Raspberry Pi, and actuators to simulate and control pond conditions in real time.
- **Validation**: Fish growth and water condition changes are observed and recorded to evaluate system effectiveness.

Team Member

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Architecture



System Architecture

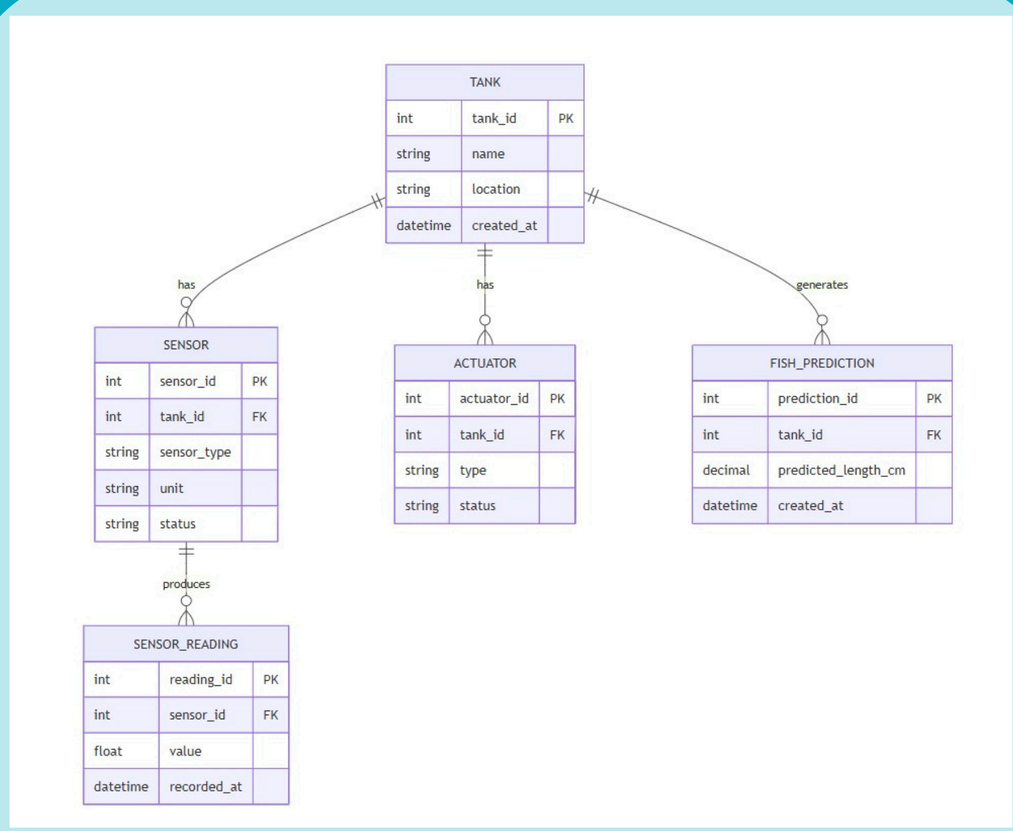
IoT Sensing (The Eyes):
Smart sensors that monitor water health metrics (pH, Oxygen, Temperature, etc) 24/7.

AI Core (The Brain):
An intelligent processor that analyses data in real-time to detect stress and decide what the pond needs.

Automated Response (The Hands):
The system instantly activates Feeders and Aerators based on the needs.

Dashboard (The View):
A simple, real-time web interface that give you an overview, status and control of the farm.

ERD



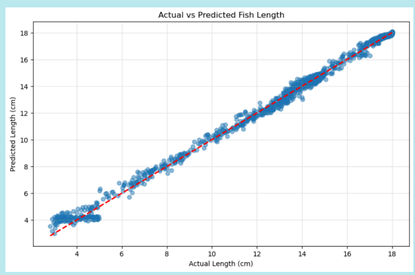
Results



The canvas to read datas from

Raw datas are uploaded to database

Cleaning, preprocessing, training



Output after running model prediction

The system successfully monitored environmental parameters such as temperature, pH, turbidity, and water level in real time.

Automated responses such as feeder activation were triggered based on predefined AI rules.

Sensor readings and system actions were logged, demonstrating the feasibility of an AI-assisted pond management system.

Conclusion

This project presents an AI-based system for monitoring, control, and growth prediction of ikan puyu. The system helps improve fish growth efficiency and supports better decision-making through data-driven monitoring in aquaculture.Overall, the system provides a practical and scalable solution for smart aquaculture.